

Week 5: grouping right coronary arteries by tortuosity

1 Low, average and high tortuosity in 560 right coronary arteries

Mean absolute curvature is the geometric measure that tracks low shear

Kashyap et al. found mean absolute curvature, not the tortuosity index, to be the best geometric descriptor of low wall shear stress (WSS) [1]. They simulated blood flow through the left main bifurcation of 127 people without coronary artery disease (CAD), took the share of the wall below 0.4 Pa of time-averaged WSS as the low-shear zone, and regressed four tortuosity measures against it: mean absolute curvature explained the most variance, $R^2 = 0.112$ ($p < 0.001$), while the tortuosity index, arc length over the straight-line distance between the ends, explained none ($p = 0.865$) [1]. Mean absolute curvature is the total angle through which the vessel turns along its length, counting left and right turns alike, divided by that length; it is measured in radians per millimeter, written mm^{-1} because a radian has no unit.

A turning-angle score is linked to CAD in 22,334 patients

Tello Ayala et al. scored the right coronary artery (RCA) in 38,691 invasive coronary angiograms from 22,334 patients referred for catheterization, using a centerline traced in a single 2D X-ray frame (Figure 1a) [2]. The centerline is a list of points p_i ordered from the start of the RCA to its distal end [2], so the direction of the vessel at each point follows from its neighbors. The step arriving at p_i is the vector $a_i = p_i - p_{i-1}$, and the step leaving it is $b_i = p_{i+1} - p_i$. The turning angle θ_i is the angle needed to rotate the arriving step onto the leaving one [2], and it follows from the dot product of the two steps,

$$\cos \theta_i = \frac{a_i \cdot b_i}{|a_i| |b_i|}, \quad (1)$$

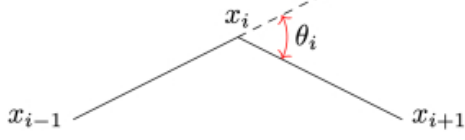
so $\theta_i = 0$ where the vessel runs straight and grows as it bends. The score is the mean turning angle over all n points, divided by π ,

$$T = \frac{1}{n} \sum_{i=1}^n \frac{|\theta_i|}{\pi}, \quad (2)$$

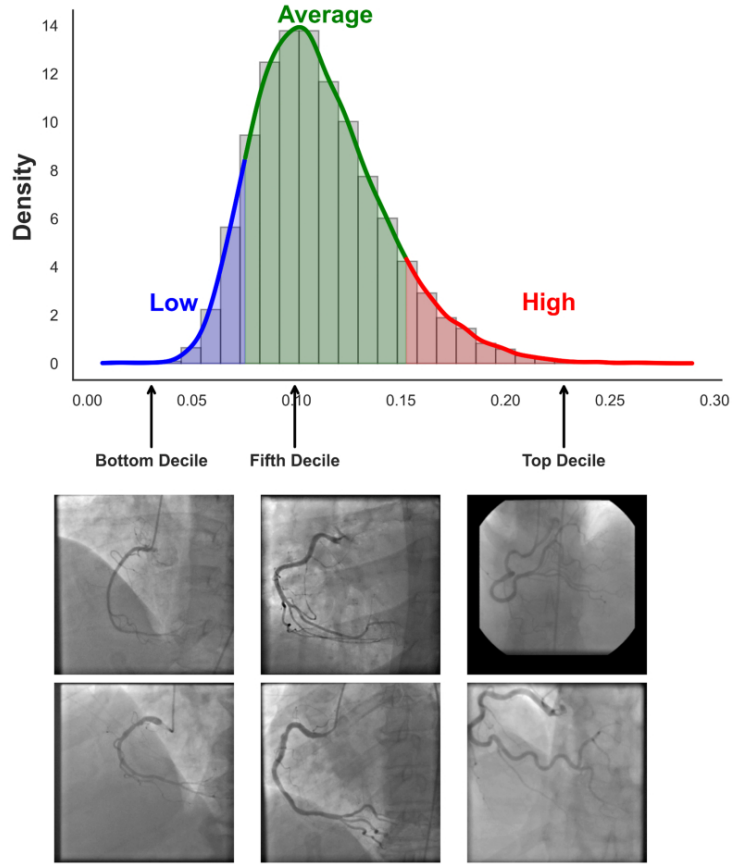
which runs from 0 for a straight line to 1 for a line that doubles back at every point [2]. They split the scores into low (bottom decile), average (middle eight deciles) and high (top decile) tortuosity (Figure 1b), and each standard deviation of T raised the adjusted odds of CAD by 5% (odds ratio 1.05, 95% CI 1.03 to 1.08) [2]. A blinded interventional cardiologist labeled 100 angiograms from each group as tortuous or not; 75% of the high group was judged tortuous, only 58% of the arteries judged tortuous were in the high group, and agreement was Cohen's $\kappa = 0.52$ (95% CI 0.41 to 0.63) [2]. With evenly spaced points, the mean turning angle is the total turning divided by the number of steps, so T measures the same quantity as mean absolute curvature [3].



B



(a) From angiogram to scored centerline; the turning angle θ_i at point x_i (called p_i in the text) is the local tortuosity.



(b) Score distribution split at the 10th and 90th percentiles, with two example angiograms per group.

Figure 1: Tello Ayala et al. score each right coronary artery by the mean turning angle along its 2D skeleton and group the scores at the 10th and 90th percentiles; their axis is a mean angle per skeleton point over π , not comparable to Figure 2. Reproduced unmodified from Tello Ayala et al. [2], JACC: Advances 2026, CC BY-NC-ND 4.0.

The mean absolute turning angle is implemented on the 3D centerline

An angiogram is an X-ray projection of the tree onto one plane. A stretch of vessel running toward or away from the X-ray source is foreshortened, so a bend in that direction shrinks or disappears, while a gentle bend seen at a slant can look sharper than it is. Tello Ayala

et al. scored the RCA only in left anterior oblique (LAO) recordings [2], so the bends they measured were those visible from that direction. Our centerline comes from a 3D image instead: ImageCAS-X delivers each coronary tree from the coronary CT angiography (CCTA) volume as a 3D centerline, a list of points (x, y, z) in millimeters about 0.47 mm apart, with every point labeled by artery [4]. The RCA is the longest path through the points labeled RCA, from the ostium to where that label ends at the crux of the heart, so side branches are removed by their labels; the posterior descending artery (PDA), which Tello Ayala et al. included, carries its own label here and is left out. I implement the mean absolute turning angle on this centerline in the same steps (`tortuosity/merged.py`), with one change: the points are first respaced a fixed distance ℓ apart. Starting at one end of the RCA, each new point v_j is placed on the centerline exactly ℓ in a straight line from the previous one, so every step is a straight chord of length ℓ , and a remainder shorter than ℓ at the far end is dropped. The points v_j are ordered along the vessel, so the direction at each point again follows from its neighbors. The step arriving at v_j is $a_j = v_j - v_{j-1}$, and the step leaving it is $b_j = v_{j+1} - v_j$, now vectors with three components (x, y, z) . The turning angle θ_j between them follows from the same dot product, Equation (1), with the z components added to the products. For example, with $\ell = 5$ mm and points $(0, 0, 0)$, $(5, 0, 0)$ and $(9.70, 1.71, 0)$, the steps are $(5, 0, 0)$ and $(4.70, 1.71, 0)$, the cosine is $23.5/25 = 0.94$, and the vessel turned 20° at the middle point. The score is the total turning divided by π times the length,

$$T_\ell = \frac{1}{\pi L_c} \sum_j \theta_j, \quad (3)$$

where L_c is the total length of the steps, their number times ℓ . In 3D the angle has no sign, which Equation (2) discards anyway. Dividing by length rather than by the number of points makes T_ℓ the mean absolute turning angle per millimeter, divided by π , in mm^{-1} ; this is the mean absolute curvature measured at scale ℓ .

The chord is 5 mm instead of the skeleton step

Tello Ayala et al. took the turning angle between neighboring skeleton points and do not report how far apart these are [2]. If that step is short, the score measures position error as much as shape. If each vertex is misplaced by a random error of standard deviation σ , each turning angle gains about $3\sigma/\ell$ radians, so the score gains about $3\sigma/\ell^2$; halving the chord quadruples the error, and because the angles enter as magnitudes it is always added, never averaged out. The ImageCAS-X RCAs show this directly: their median score falls from 0.027 mm^{-1} at $\ell = 2$ mm to 0.015 mm^{-1} at 5 mm, a 44% drop, but only to 0.013 mm^{-1} at 8 mm. This thesis uses $\ell = 5$ mm, the scale van Zandwijk et al. chose for coronary curvature on CCTA in 71 patients as “an adequate and robust option” [5]. Their mean artery curvature was 0.081 to 0.090 mm^{-1} across the cardiac cycle [5], while our median RCA turns 0.048 radians per millimeter, about half, for reasons not yet known. A 5 mm chord cannot see bends shorter than a few millimeters, which is the price of the lower noise. The delivered centerlines are already smoothed [4], so no further smoothing is applied.

Results: the groups on 560 ImageCAS-X RCAs

All 560 RCAs in the ImageCAS-X training split were scored; none was shorter than the 15 mm minimum, and their median length was 106 mm (5th to 95th percentile 69 to 137 mm). The median T_5 was 0.0152 mm^{-1} (interquartile range 0.0127 to 0.0190), about 2.7° of turning per millimeter, and the low and high groups begin below 0.0115 and above 0.0237 mm^{-1} , giving 56, 448 and 56 RCAs (Figure 2). The high cut-off is twice the low one: 4.3° against 2.1° of turning per millimeter.

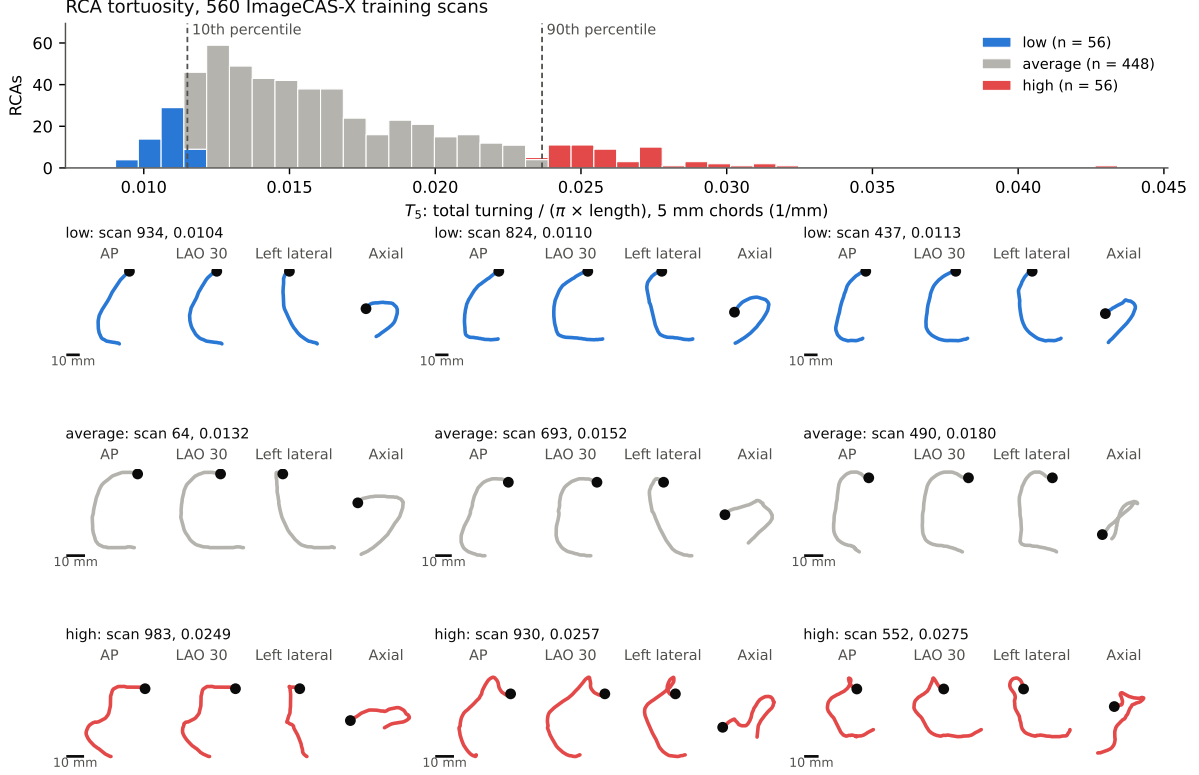


Figure 2: The three top-decile right coronary arteries bend more often and more tightly than the three bottom-decile ones, in every view. Top: T_5 for 560 ImageCAS-X training scans, split at the 10th and 90th percentiles as in Tello Ayala et al. [2]. Bottom: three RCAs per group, at the group’s 25th, 50th and 75th percentile, drawn in four projections that share one scale per RCA (see the 10 mm bar); the dot marks the ostium.

The top decile is stable to the choice of chord length and the bottom decile is less so. Rank correlation between T_5 and the same score at 4 and 8 mm is 0.97 and 0.96, yet at 4 mm 7 of the 56 high RCAs and 14 of the 56 low RCAs change group, and at 8 mm 9 and 18 do. Longer RCAs score slightly higher (Spearman $\rho = 0.30$ with length); the high group has a median length of 118 mm against 103 and 105 mm in the other two, so length is reported beside every group comparison. In two of the three high examples, scans 930 and 552, the tightest bend is a loop at the ostium; whether the high group is driven by the take-off or by bends along the vessel is open. Our numbers cannot be compared to the 90th percentile of 0.152 reported by Tello Ayala et al. [2]: theirs is a mean angle per skeleton step and ours per 5 mm chord (0.118π , or 21° , at our 90th percentile), and the angle grows with the step.

Next: a blinded visual check

To repeat the check of Tello Ayala et al., 50 RCAs were drawn from each group (seed 42), rendered in the four views of Figure 2 with no score shown, and shuffled. I will label each tortuous or not, then relabel 20 of them at least a week later, since my agreement with myself bounds any agreement with the score. Agreement is reported as precision, recall and Cohen’s κ with a bootstrap 95% CI, next to their $\kappa = 0.52$. At $\kappa \approx 0.5$, 150 labels give an interval of about ± 0.14 , narrow enough to tell whether the 3D score agrees with the eye better or worse than the 2D score did.

	Tello Ayala et al. [2]	This work
Images	invasive angiography, LAO view	CCTA
Centerline	U-Net outline of one 2D frame, skeletonized	delivered 3D centerline, labeled by artery
RCA extent	ostium to distal end of PDA	ostium to end of the RCA label at the crux
Cohort	22,334 patients referred for catheterization	560 ImageCAS-X training scans
Turning angle	between consecutive skeleton points	between 5 mm chords
Normalization	mean over points, divided by π	sum over length, divided by π (1/mm)
Smoothing	not reported	none beyond the delivered centerline
Visual check	1 interventional cardiologist, 100 per group	1 student rater, 50 per group, 20 repeats

Table 1: Every departure from Tello Ayala et al. that a comparison with their numbers must state.

References

- [1] Vikas Kashyap et al. “Accuracy of vascular tortuosity measures using computational modelling”. In: *Scientific Reports* 12.1 (2022), p. 865. DOI: [10.1038/s41598-022-04796-w](https://doi.org/10.1038/s41598-022-04796-w).
- [2] J. R. Tello Ayala, K. Supriami, S. Swaroop, et al. “A Machine Learning Driven Approach to Quantifying Coronary Artery Tortuosity”. In: *JACC: Advances* 5.6 (2026), p. 102829. DOI: [10.1016/j.jacadv.2026.102829](https://doi.org/10.1016/j.jacadv.2026.102829).
- [3] Elizabeth Bullitt et al. “Measuring tortuosity of the intracerebral vasculature from MRA images”. In: *IEEE Transactions on Medical Imaging* 22.9 (2003), pp. 1163–1171. DOI: [10.1109/TMI.2003.816964](https://doi.org/10.1109/TMI.2003.816964).
- [4] Kit M. Bransby et al. “ImageCAS-X: A dataset and benchmark for validating coronary vessel segmentation and centerline extraction in computed tomography angiography”. In: *arXiv preprint* (2026).
- [5] Jordy K. van Zandwijk et al. “Assessment of Dynamic Change of Coronary Artery Geometry and Its Relationship to Coronary Artery Disease, Based on Coronary CT Angiography”. In: *Journal of Digital Imaging* 33.2 (2020), pp. 480–489. DOI: [10.1007/s10278-019-00300-5](https://doi.org/10.1007/s10278-019-00300-5).